

a) Area Tolerance: Geometric Scaling

For a planar convex region Ω with region Ω_r generated by applying boundary perturbation r :

$$A(\Omega_r) = A(\Omega) + L(\partial\Omega)r + \pi r^2$$

$$\Delta A = A(\Omega_r) - A(\Omega) = L(\partial\Omega)r + \pi r^2$$

For small r :

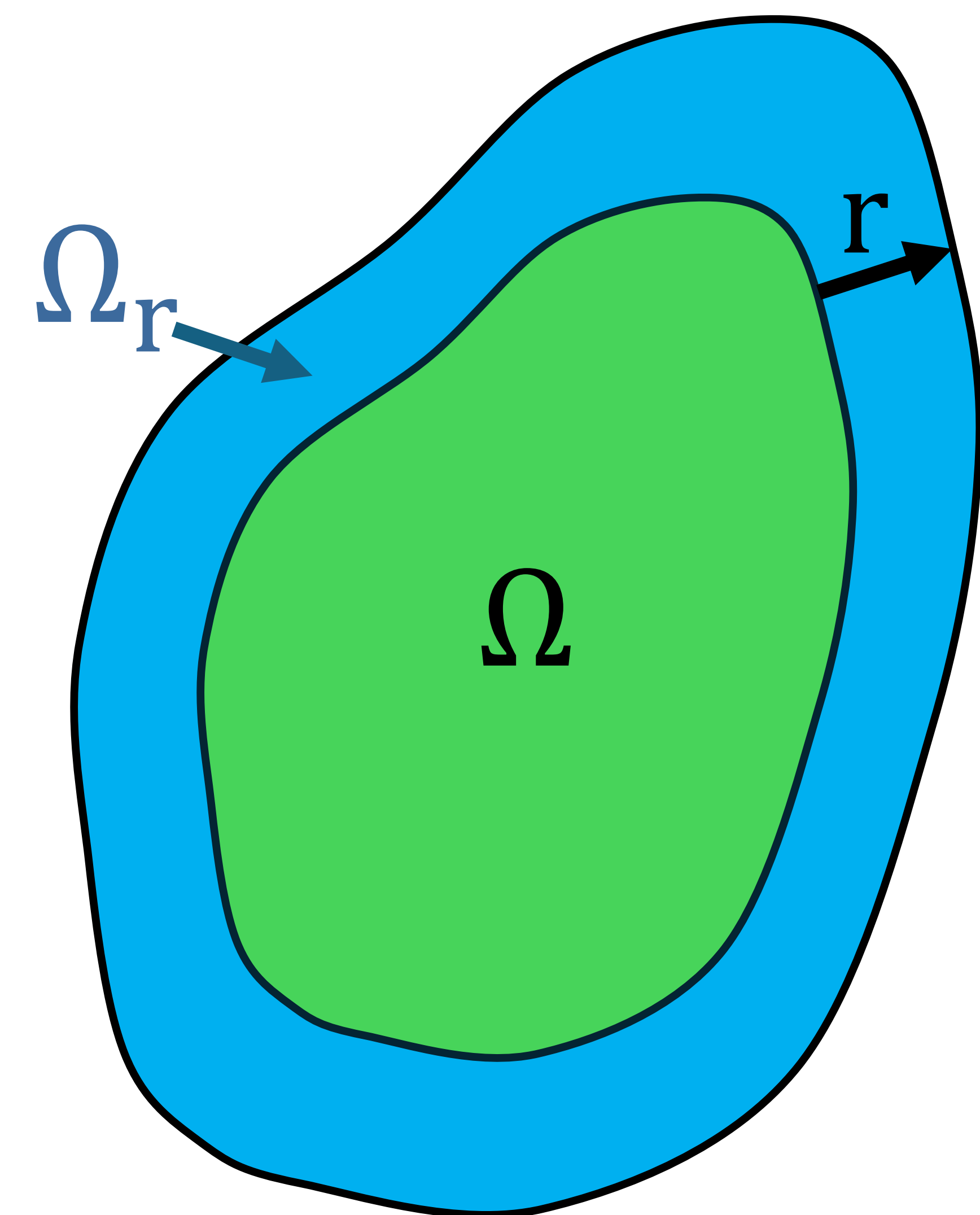
$$\Delta A \approx L(\partial\Omega)r = Pr$$

Since Ω and Ω_r exist within the same shape family:

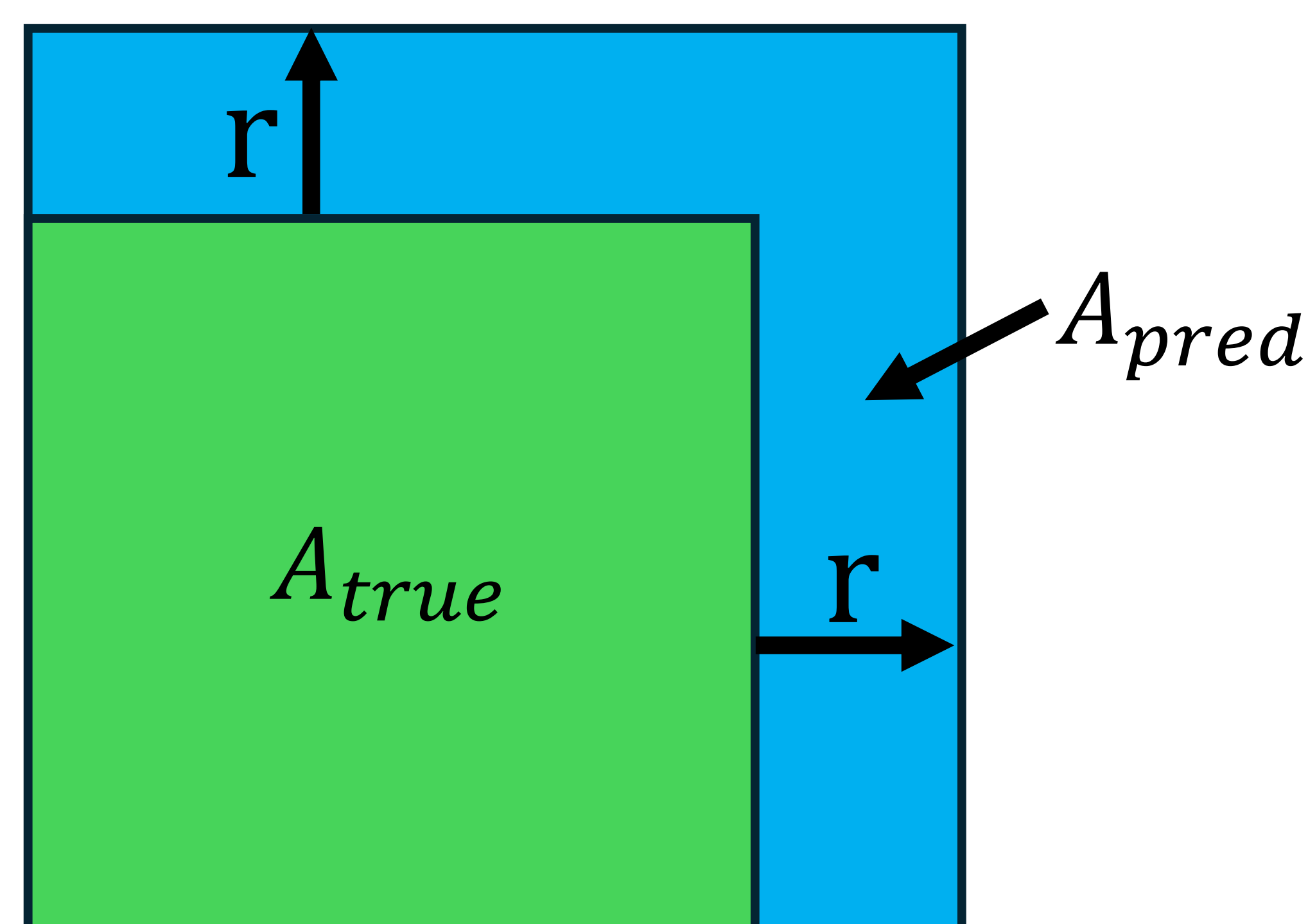
$$P \propto \sqrt{A}$$

Therefore,

$$\Delta A \propto \sqrt{A} \cdot r$$



b) Area Tolerance: Rectangular Example



$$A_{true} = L^2$$

$$A_{pred} = (L + 2r)^2$$

$$\Delta A = A_{pred} - A_{true} = 4Lr + 4r^2 \approx 4\sqrt{A_{true}}r$$

$$\Delta A \propto \sqrt{A_{true}} \Leftrightarrow r \approx \text{constant}$$

$$\text{Chosen tolerance} \Rightarrow r \approx 0.25 - 0.3 \text{ cm}$$